

ORAVITA, Romania

WMO: 153380

Lat: 45.0386N

Lon: 21.7106E

Elev: 1018

StdP: 14.16

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	19.5	8.3	8.7	20.9	12.6	10.8	24.1	45.7	31.3	35.7	33.5	3.5	320	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.1	91.1	68.5	87.6	68.9	84.5	68.4	72.7	84.4	71.0	82.8	69.5	81.4	6.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.6	109.0	79.5	66.9	102.9	77.6	65.5	97.9	75.5	36.9	84.4	35.5	82.5	34.2	81.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
34.7	23.1	18.2	8.6	94.2	5.4	4.5	4.7	97.4	1.5	100.0	-1.5	102.5	-5.5	105.7
			7.8	75.5	4.9	2.9	4.3	77.6	1.4	79.3	-1.3	80.9	-4.9	83.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	34.0	38.8	44.3	54.0	62.1	68.7	72.0	72.6	63.9	55.1	46.0	36.3	(6)
(7)		DBStd	15.82	9.34	9.22	8.47	9.16	7.28	6.57	6.31	7.29	8.39	9.11	9.91	8.82	(7)
(8)		HDD50	1794	501	327	216	65	4	0	0	2	55	191	433	(8)	
(9)		HDD65	4797	961	734	644	344	142	38	13	17	121	323	570	890	(9)
(10)		CDD50	3276	5	13	38	185	379	560	683	700	417	215	73	8	(10)
(11)		CDD65	803	0	0	1	13	52	148	231	252	87	18	1	0	(11)
(12)		CDH74	6269	0	0	1	50	265	1103	1964	2199	626	58	3	0	(12)
(13)		CDH80	2017	0	0	0	1	24	288	686	833	182	3	0	0	(13)
(14)	Wind	WSAvg	5.8	5.3	6.1	6.4	7.3	6.2	4.5	4.5	5.3	6.2	6.0	6.7	5.1	(14)
(15)	Precipitation	PrecAvg	30.2	1.9	1.7	1.7	2.6	3.1	3.6	3.8	2.3	2.8	2.4	2.0	2.2	(15)
(16)		PrecMax	39.2	3.8	4.3	4.7	4.7	6.1	8.1	8.7	4.8	7.4	5.2	5.5	5.7	(16)
(17)		PrecMin	17.4	0.5	0.1	0.5	0.9	0.9	0.4	0.3	0.3	0.5	0.1	0.0	0.4	(17)
(18)		PrecStd	6.0	0.9	1.1	1.1	1.0	1.3	2.0	2.3	1.3	1.6	1.3	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.2	63.7	70.4	78.9	82.7	89.3	94.4	95.5	91.1	79.4	70.5	60.7	(19)
(20)			MCWB	48.3	50.5	54.7	59.8	66.2	70.6	68.5	69.7	66.6	61.2	55.6	50.1	(20)
(21)		2%	DB	54.9	59.0	65.4	74.7	79.8	86.1	90.5	91.8	84.6	75.0	65.8	56.5	(21)
(22)			MCWB	45.8	48.5	51.2	57.8	64.5	70.0	68.9	68.3	66.4	60.5	54.5	47.7	(22)
(23)		5%	DB	51.1	55.9	61.7	71.0	77.0	83.3	86.7	88.2	80.6	71.4	63.5	52.8	(23)
(24)			MCWB	43.7	46.9	49.4	55.9	62.9	68.9	69.2	68.7	64.4	58.6	53.8	45.3	(24)
(25)		10%	DB	47.7	52.4	57.6	67.5	74.1	80.6	83.6	84.4	77.0	68.1	60.2	48.5	(25)
(26)			MCWB	41.3	44.4	47.3	54.3	61.2	67.7	68.6	68.0	63.2	57.9	51.8	42.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	51.9	55.4	62.1	68.4	74.7	74.5	74.6	70.3	66.0	56.5	51.8	(27)
(28)			MCD B	58.0	61.2	70.3	75.3	81.0	85.1	86.5	88.3	85.4	75.0	66.7	57.1	(28)
(29)		2%	WB	46.8	49.3	52.1	59.4	66.2	72.2	72.4	72.2	67.6	62.2	55.4	48.4	(29)
(30)			MCD B	53.9	57.6	63.8	71.7	77.9	82.1	84.1	85.1	82.8	70.3	64.9	56.1	(30)
(31)		5%	WB	44.4	47.4	50.1	57.1	64.1	70.0	70.8	70.5	65.8	60.5	54.2	45.5	(31)
(32)			MCD B	50.2	54.8	60.4	68.8	75.0	80.6	82.5	83.3	77.5	69.2	62.6	51.7	(32)
(33)		10%	WB	42.1	45.2	47.9	55.2	62.0	68.5	69.3	68.9	63.9	58.5	52.5	43.4	(33)
(34)			MCD B	46.8	51.9	56.4	65.8	71.9	78.8	80.8	81.2	74.2	66.9	59.7	48.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	12.4	14.3	15.1	15.6	16.6	17.6	17.1	15.0	13.6	10.8	9.6	(35)
(36)			MCDBR	14.7	16.8	19.3	18.2	17.8	19.3	20.4	20.1	18.3	17.1	14.1	14.3	(36)
(37)			MCWBR	8.8	9.6	10.6	9.2	9.3	9.5	8.8	7.8	7.7	8.5	7.0	8.5	(37)
(38)		5% WB	MCD BR	13.4	15.4	17.5	17.9	17.0	18.9	18.6	18.9	16.6	15.7	13.1	12.9	(38)
(39)			MCWBR	8.2	9.3	10.5	9.7	9.6	10.1	9.7	9.0	8.0	8.4	6.9	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.333	0.361	0.399	0.455	0.441	0.433	0.442	0.457	0.427	0.387	0.355	0.331	(40)	
(41)		taud	2.313	2.225	2.197	2.061	2.161	2.223	2.199	2.155	2.206	2.314	2.355	2.347	(41)	
(42)		Ebn at Noon	241	255	261	256	264	266	262	252	249	243	232	229	(42)	
(43)		Edh at Noon	26	34	40	49	46	44	44	45	39	31	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	391	654	1042	1442	1743	1942	1947	1737	1217	827	495	326	(44)	
(45)		RadStd	48	87	136	149	166	180	150	167	128	96	50	47	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (62 neighbors)	+1.17	N/A	N/A	N/A	+0.65	N/A	N/A	-283	+196	+108	(47)

Nomenclature: See separate page